Resolved Physical Direction, Response Degeneracy, and Observable-Tensor Automation on Real LIGO H1–L1 Backgrounds

Matěj Rada

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Abstract

We test whether a response-relative Resolution Geometry (RG) pipeline can distinguish recoverable physical direction, near-null instability, and exact observational degeneracy while operating on real LIGO Hanford H1 and Livingston L1 strain backgrounds. Two orthonormal, quadrature chirp coordinates define a controlled two-dimensional latent source. The same deterministic latent source sequence is projected into 256 four-second windows drawn across each independent 4096 s O4a detector record. Three declared two-detector response operators are attacked: a well-conditioned full-rank operator, a near-degenerate full-rank operator, and an exactly rank-one operator. For the well-conditioned operator, increasing injection level from 1 to 8 decreases median resolved-direction error from 0.5734 to 0.08219 rad and resolved-tensor relative error from 2.4327 to 0.07916. For the near-degenerate operator, whose condition number is 85.35, direction and tensor reconstruction remain unstable even at the strongest tested injection; at level 8 the median direction error is 0.9897 rad and tensor error is 54.20. For the rank-one operator, large latent perturbations in the exact response kernel alter sensor observations only at relative magnitude $1.4–1.6 \times 10^{-16}$, while the observable rank-one tensor converges to relative error 0.02002 at injection level 8. A singular-threshold sweep correctly changes the near-degenerate resolved rank from one to two when the threshold passes below the second relative singular value. The computation therefore supports response-relative recovery of resolved source direction, explicit preservation of null equivalence classes, and automated second-moment tensors on the observable source sector. It does not establish that the declared response matrices are the physical H1–L1 antenna response, nor does it constitute astrophysical source reconstruction or event detection.

1 Problem and Logical Levels

The purpose of this study is to close a specific gap left by detector-local observable extraction. Covariance geometry computed from detector summaries can establish empirical resolved sectors, but it does not by itself attach those sectors to a known physical source direction. Conversely, a controlled projection with known latent source and known response permits three logically distinct questions:

1. which latent source directions are observable through the declared response;
2. which source changes are observationally degenerate at a stated resolution;
3. whether an observable rank-two source tensor can be constructed automatically and recovered under real detector backgrounds.

The present battery answers these questions for controlled software injections into authentic H1 and L1 strain backgrounds. The source and response are declared experimental objects; the detector noise is real. No claim is made that the controlled response equals the calibrated astrophysical antenna response.

2 Data and Controlled Source

The backgrounds are independent 4096 s O4a strain records sampled at 16384 Hz:

- H1: `H-H1_GWOSC_O4a_16KHZ_R1-1368424448-4096.hdf5`;
- L1: `L-L1_GWOSC_O4a_16KHZ_R1-1368977408-4096.hdf5`.

From each record, 256 four-second windows are selected at evenly distributed indices across the full 4096 s interval. Each window contains 65536 strain samples. The detector records are noncontemporaneous and are used only as independent real noise backgrounds.

Let $q_1, q_2 \in \mathbb{R}^{65536}$ be orthonormal quadrature chirp templates with a sinusoidal taper and instantaneous frequency increasing from 35 Hz to 350 Hz over four seconds. The controlled latent source coordinate for window j is

$$s_j = (s_{j1}, s_{j2}) \in \mathbb{R}^2,$$

where the 256 source vectors are drawn from a fixed-seed standard normal sequence and standardized coordinatewise. The same sequence $\{s_j\}$ is used for H1 and L1.

For detector D, projection of the real strain window $h_{D,j}$ onto the template pair produces a two-coordinate background vector. The first detector-local matched-filter coordinate is robustly standardized using its median and median absolute deviation. The two-detector observation is then

$$y_j = n_j + \alpha A s_j,$$

where $n_j \in \mathbb{R}^2$ consists of the standardized H1 and L1 background coordinates, $A \in \mathbb{R}^{2\times 2}$ is a declared response matrix, and $\alpha > 0$ is the tested injection level.

Remark 2.1 (Meaning of injection level)**.** *The parameter α is a dimensionless multiplier in detector-locally standardized matched-filter coordinates. It should not be identified with an astrophysical signal-to-noise ratio without a calibrated noise-power and response model.*

3 Response-Relative Resolution Geometry

Definition 3.1 (Resolved and null source sectors)**.** *Let*

$$A = U\Sigma V^{\mathsf{T}}, \qquad \Sigma = \operatorname{diag}(\sigma_1, \sigma_2), \qquad \sigma_1 \geq \sigma_2 \geq 0.$$

For relative response threshold $\epsilon > 0$, define

$$\mathcal{R}_\epsilon(A) = \operatorname{span}\{v_i : \sigma_i/\sigma_1 > \epsilon\}, \qquad \mathcal{N}_\epsilon(A) = \mathcal{R}_\epsilon(A)^\perp,$$

and let $P_{R,\epsilon}$ and $P_{N,\epsilon}$ be the corresponding orthogonal projectors.

Definition 3.2 (Observable source equivalence)**.** *Two latent sources $s, s' \in \mathbb{R}^2$ are observationally equivalent at resolution ϵ when*

$$P_{R,\epsilon} s = P_{R,\epsilon} s'.$$

For an exact kernel this reduces to

$$s' \sim_A s \quad \Longleftrightarrow \quad s' - s \in \ker A.$$

The observable object is the equivalence class $[s]_{\mathrm{obs}} = s + \mathcal{N}_\epsilon(A)$, not an arbitrary full-source representative.

Definition 3.3 (Resolved reconstruction). *Given $y_j = n_j + \alpha A s_j$, the unregularized controlled reconstruction is*

$$\widehat{s}_j = \frac{1}{\alpha} A^{+} y_j,$$

where A^{+} is the Moore–Penrose pseudoinverse. The true and reconstructed resolved sources are

$$s_{R,j} = P_R s_j, \qquad \widehat{s}_{R,j} = P_R \widehat{s}_j.$$

Only these projected quantities are compared.

Definition 3.4 (Observable second-moment tensor). *For m source windows, define*

$$Q_R = \frac{1}{m-1} \sum_{j=1}^{m} (s_{R,j} - \bar{s}_R)(s_{R,j} - \bar{s}_R)^{\mathsf{T}},$$

and

$$\widehat{Q}_R = \frac{1}{m-1} \sum_{j=1}^{m} (\widehat{s}_{R,j} - \bar{\widehat{s}}_R)(\widehat{s}_{R,j} - \bar{\widehat{s}}_R)^{\mathsf{T}}.$$

The tensor error is

$$E_Q = \frac{\|\widehat{Q}_R - Q_R\|_F}{\|Q_R\|_F}.$$

The direction score for each nonzero resolved source pair is

$$\theta_j = \arccos\left(\frac{\langle \widehat{s}_{R,j}, s_{R,j} \rangle}{\|\widehat{s}_{R,j}\| \, \|s_{R,j}\|} \right).$$

Signed direction is retained; no absolute value is inserted into the inner product.

4 Response Operators

Three operators isolate three distinct regimes:

$$A_{\text{good}} = \begin{pmatrix} 1 & 0.20 \\ 0.30 & 0.90 \end{pmatrix}, \qquad \sigma(A_{\text{good}}) = (1.2063, 0.6964), \qquad \kappa = 1.7322, \tag{1}$$

$$A_{\text{near}} = \begin{pmatrix} 1 & 0.20 \\ 0.98 & 0.22 \end{pmatrix}, \qquad \sigma(A_{\text{near}}) = (1.4313, 0.01677), \qquad \kappa = 85.355, \tag{2}$$

$$A_{\text{null}} = \begin{pmatrix} 1 & 0.20 \\ 2 & 0.40 \end{pmatrix}, \qquad \sigma(A_{\text{null}}) = (2.2804, 5.80 \times 10^{-17}), \qquad \text{rank} = 1. \tag{3}$$

The first permits stable inversion at sufficient injection strength. The second is algebraically full rank but strongly amplifies background noise in its weak direction. The third has an exact one-dimensional source kernel.

5 Theorem Ladder

Theorem 5.1 (Exact response-kernel equivalence). *Let $A : \mathbb{R}^p \to \mathbb{R}^m$ be linear. If $n \in \ker A$, then for every source s and scalar β,*

$$A(s + \beta n) = As.$$

Consequently, no reconstruction based only on As can distinguish s from $s + \beta n$.

Proof. Linearity gives

$$A(s + \beta n) = As + \beta An = As,$$

because $An = 0$. Therefore the observation factors through the quotient $\mathbb{R}^p / \ker A$. $\square$

Corollary 5.1 (Tensor identifiability is sector-relative)**.** *A source second-moment tensor is identifiable from the response only after projection onto the resolved quotient. In general,* $\mathrm{Cov}(s)$ *is not identifiable, whereas*

$$Q_R = \mathrm{Cov}(P_R s)$$

is the appropriate observable tensor target.

Proof. Null components may be changed arbitrarily without changing the observation. Such changes can alter the full covariance $\mathrm{Cov}(s)$ while leaving As fixed. Projection removes precisely those unresolved components. $\square$

Proposition 5.1 (Noise amplification by weak response directions)**.** *For the reconstruction*

$$\widehat{s} - s = A^{+} n / \alpha,$$

the error component along right singular vector v_i is scaled by $1/(\alpha\sigma_i)$. Hence weak nonzero singular directions can be less recoverable than stronger directions even though the algebraic rank is unchanged.

Proof. Using $A^{+} = V\Sigma^{+}U^{\mathsf{T}}$ gives

$$\widehat{s} - s = \frac{1}{\alpha} \sum_{\sigma_i > 0} \frac{u_i^{\mathsf{T}} n}{\sigma_i} v_i.$$

The displayed coefficient proves the claim. $\square$

Corollary 5.2 (Resolution gate precedes physical interpretation)**.** *Algebraic full rank alone does not justify a physical-direction claim. A direction should be admitted only with a declared threshold or uncertainty rule that controls amplification by $1/\sigma_i$.*

Theorem 5.2 (Consistency of resolved source and tensor reconstruction)**.** *Fix a response matrix A and its resolved projector P_R. Suppose the background satisfies*

$$\frac{1}{\alpha} P_R A^{+} n_j \longrightarrow 0$$

in mean square as the effective injection strength increases. Then

$$\widehat{s}_{R,j} \longrightarrow s_{R,j}$$

in mean square. For a fixed finite sample with bounded moments, the empirical observable tensor satisfies

$$\widehat{Q}_R \longrightarrow Q_R.$$

tlThe conclusion concerns the resolved sector only.

Proof. Projection of the reconstruction identity gives

$$\widehat{s}_{R,j} - s_{R,j} = \frac{1}{\alpha} P_R A^{+} n_j.$$

The assumed mean-square convergence proves source consistency. The empirical covariance is a continuous polynomial function of the finitely many reconstructed source coordinates, so convergence of those coordinates implies convergence of the empirical covariance tensor. $\square$

Remark 5.1 (Computational status)**.** *The exact-kernel theorem and noise-amplification proposition are analytic. The numerical results below are finite-sample evidence that their predicted regimes occur on the selected real H1 and L1 backgrounds.*

6 Well-Conditioned Direction and Tensor Recovery

Proposition 6.1 (Observed recovery under A_{good}). *For the well-conditioned response, increasing injection level from 1 to 8 decreases the median resolved-direction error from* 0.57336 *to* 0.08219 *rad, resolved-source relative error from* 1.3555 *to* 0.16943*, and resolved-tensor relative error from* 2.4327 *to* 0.07916*.*

Proof. These values are obtained by direct reconstruction of the fixed latent sequence from the controlled projections into the selected real detector backgrounds. □

Table 1: Well-conditioned response recovery.

Injection level	Median direction	90th percentile direction	Source error	Tensor error
1	0.57336	1.67007	1.35547	2.43269
2	0.33431	1.17265	0.67773	0.70219
4	0.17356	0.67132	0.33887	0.22257
8	0.08219	0.34760	0.16943	0.07916

Corollary 6.1 (Finite-background physical-direction support). *Within the declared controlled response model, the battery supports recovery of a known resolved physical source direction on real H1/L1 backgrounds when conditioning and injection strength are sufficient.*

This corollary is deliberately narrower than an astrophysical direction claim. It validates the projection-and-recovery mechanism under a known operator; it does not identify a sky location, polarization, or actual gravitational-wave source.

7 Near-Degenerate Failure and Threshold Crossing

Proposition 7.1 (Observed instability under A_{near}). *For the near-degenerate response with condition number* 85.35*, reconstruction remains unstable throughout the tested range. At injection level 8, the median direction error is* 0.98968 *rad, source relative error is* 6.1012*, and tensor relative error is* 54.199*.*

Proof. Direct pseudoinverse reconstruction gives the values in Table 2. The errors decrease with injection level, but remain far above the well-conditioned errors because the weak singular direction amplifies detector background. □

Table 2: Near-degenerate response recovery.

Injection level	Median direction	90th percentile direction	Source error	Tensor error
1	1.38238	2.56408	48.8095	3370.897
2	1.26685	2.40464	24.4048	846.216
4	1.14394	2.22409	12.2024	213.301
8	0.98968	2.06084	6.10119	54.1992

The relative second singular value is

$$\frac{\sigma_2}{\sigma_1} = 0.011715\ldots.$$

The strict resolution gate therefore gives

$$\operatorname{rank}_{0.1} A_{\text{near}} = ank_{0.05} A_{\text{near}} = 1,$$

and

$$\operatorname{rank}_{0.01} A_{\text{near}} = ank_{0.005} A_{\text{near}} = ank_{0.001} A_{\text{near}} = 2.$$

Corollary 7.1 (Algebraic rank is not an admissibility certificate)**.** *The near-degenerate computation rejects the implication*

$$\operatorname{rank} A = p \quad \Longrightarrow \quad \textit{all source directions are stably recoverable.}$$

Numerical admissibility must include conditioning or uncertainty, not rank alone.

8 Exact Null Injection and Observable Equivalence

For A_{null}, let n_0 be a unit vector spanning $\ker A_{\text{null}}$. For every source window, an independent perturbation

$$s'_j = s_j + \beta_j n_0, \qquad \beta_j \sim 3\mathcal{N}(0,1),$$

is injected into the latent source.

Proposition 8.1 (Observed null invariance)**.** *Across the five tested injection levels, adding the large latent null component changes the response by relative norm between* 1.39×10^{-16} *and* 1.57×10^{-16}*. The projected resolved source is unchanged to numerical precision.*

Proof. The response difference is computed directly as

$$(s'_j - s_j) A_{\text{null}}^{\mathsf{T}}.$$

Because the perturbation lies in the exact kernel, the analytic difference vanishes; the reported nonzero values are floating-point roundoff. □

Table 3: Rank-one resolved reconstruction and null injection.

Injection level	Source error	Tensor error	Relative sensor change	Direction error
1	0.47544	0.35788	1.56×10^{-16}	0
2	0.23772	0.12246	1.49×10^{-16}	0
4	0.11886	0.04711	1.39×10^{-16}	0
8	0.05943	0.02002	1.57×10^{-16}	0

Remark 8.1 (One-dimensional angle)**.** *The zero direction error in Table 3 does not mean that the full two-dimensional source is recovered. Both the true and reconstructed resolved sources lie on the same one-dimensional observable axis. The unresolved orthogonal component remains arbitrary.*

Corollary 8.1 (Computational realization of the quotient source)**.** *The rank-one experiment realizes the observable object*

$$[s]_{\text{obs}} = s + \ker A_{\text{null}}$$

on real detector backgrounds: large changes of representative inside the equivalence class leave the observation unchanged.

9 Tensor-Automation Ladder

The computation supports the following response-relative automation:

$$(A, y, \epsilon) \longrightarrow (\mathcal{R}_\epsilon, \mathcal{N}_\epsilon) \longrightarrow \widehat{s}_R \longrightarrow \widehat{Q}_R.$$

It is important that the tensor be constructed after null elimination. Otherwise arbitrary variance in an unobservable source direction would be presented as if it were physically inferred.

Theorem 9.1 (Well-defined observable tensor on equivalence classes)**.** *Let $[s_j]$ be source equivalence classes modulo $\mathcal{N}_\epsilon(A)$. The assignment*

$$\{[s_j]\}_{j=1}^m \longmapsto \mathrm{Cov}(P_{R,\epsilon} s_j)$$

is independent of the representatives chosen from each class.

Proof. If $s_j' = s_j + n_j$ with $n_j \in \mathcal{N}_\epsilon(A)$, then

$$P_{R,\epsilon} s_j' = P_{R,\epsilon} s_j + P_{R,\epsilon} n_j = P_{R,\epsilon} s_j.$$

Therefore all projected samples, their mean, and their covariance are unchanged. □

Corollary 9.1 (Null-safe tensor automation)**.** *An automated tensor built from $P_R \widehat{s}$ is invariant under arbitrary changes of latent source representative within the declared null equivalence class.*

The numerical battery supplies two complementary witnesses. Under A_{good}, the full resolved tensor converges as background influence decreases. Under A_{null}, only the rank-one tensor is observable, and its reconstruction converges while arbitrary exact-null variance remains excluded.

10 What Is Established

The analytic and computational levels support the following statements:

1. A declared response operator canonically separates resolved and null source sectors at a stated threshold.

2. Exact null perturbations define genuine observational equivalence classes.

3. Weak nonzero singular directions amplify real detector background and can invalidate direction and tensor recovery despite full algebraic rank.

4. Under adequate conditioning, resolved source direction and the resolved second-moment tensor converge toward known injected truth as injection strength increases.

5. The observable tensor is well defined on source equivalence classes and must be formed after null projection.

6. Real H1/L1 backgrounds provide a hostile, non-Gaussian measurement environment in which the controlled mechanisms can be tested reproducibly.

11 Non-Claims

This study does not claim:

1. detection of a gravitational wave or identification of an astrophysical event;
2. reconstruction of sky direction, polarization, luminosity distance, waveform parameters, or source stress–energy;
3. that A_{good}, A_{near}, or A_{null} is the actual calibrated H1–L1 antenna-response matrix;
4. that the dimensionless injection multiplier is a calibrated network signal-to-noise ratio;
5. that pseudoinverse reconstruction is optimal for LIGO parameter estimation;
6. that the real H1 and L1 records are contemporaneous or contain a common physical signal;
7. that a two-dimensional chirp-quadrature source spans the full gravitational-wave source space;
8. that finite-sample convergence in this battery proves universal RG validity;
9. that a source tensor reconstructed under the controlled response equals the Einstein tensor or a physical stress–energy tensor.

The result is a controlled projection theorem-and-battery study: it validates response-relative direction, degeneracy, and tensor mechanisms under real detector backgrounds.

12 Limitations

The latent source is two-dimensional, the source distribution is synthetic, and only one fixed chirp family is used. The background projection uses one robustly standardized matched-filter coordinate per detector. The declared response matrices are pedagogical stress operators rather than calibrated antenna responses. The finite collection of 256 windows samples, but does not exhaust, detector nonstationarity. The inversion is unregularized apart from exact pseudoinverse truncation; the near-degenerate failure could be traded for bias using regularization, but regularization would not create information in an exact null sector. Finally, the strongest physical validation would use documented common software or hardware injections passed through calibrated H1/L1 responses.

13 Reproducibility

The complete numerical outputs are stored in

- `rg_h1_l1_injection_projection_results.json`,
- `rg_h1_l1_injection_projection_summary.txt`, and
- `run_rg_h1_l1_injection_projection_attack.py`.

The script records SHA-256 checksums of both input HDF5 files, the selected window indices, source seed `31072026`, sample rate, duration, response matrices, singular values, tested injection levels, direction errors, source errors, tensor errors, sensor residuals, null-injection changes, and threshold sweep. All source vectors and response operators are generated deterministically from the script and fixed seed.

The input files are

- `H-H1_GWOSC_O4a_16KHZ_R1-1368424448-4096.hdf5`,
- `L-L1_GWOSC_O4a_16KHZ_R1-1368977408-4096.hdf5`.

14 Conclusion

The battery separates three regimes that an observable-source framework must not conflate. A well-conditioned response permits increasingly accurate recovery of known source direction and its rank-two second-moment tensor on authentic detector backgrounds. A near-degenerate response is algebraically invertible but physically unreliable at the tested noise and injection levels; the singular threshold correctly exposes the change in admissible dimension. An exact rank-one response admits a large latent null component that is observationally invisible to numerical precision, forcing the source object to be an equivalence class and the tensor object to be constructed only on the resolved quotient.

The resulting supported chain is

$$\boxed{(A, y, \epsilon) \longrightarrow [s]_{\mathcal{N}_\epsilon} \longrightarrow \widehat{s}_R \longrightarrow \widehat{Q}_R}$$

under a declared response operator and real H1/L1 backgrounds. This closes the controlled computational gap between observable-sector extraction and null-safe tensor automation. The remaining physical gap is explicit rather than hidden: replacement of the declared stress operators by calibrated detector responses and documented common injections or events.